Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	AssetFlow – AI-Powered Enterprise IT Asset Management Platform with Intelligent Asset Tracking & Lifecycle Monitoring	
S. No.	**University ID**	**Name**
1	2420030249	Anakala Sarayu
2	2420030269	Manduri Varshitha
3	2420030468	Nidumolu Veda Samhita
4	2420090076	Sushmita Sallam
Name of the Guide		Ms.G Lavanya

Abstract

AssetFlow – Enterprise IT Asset Management Platform with Intelligent Asset Tracking & Lifecycle Monitoring is a modern, centralized web-based application developed to simplify and automate the management of an organization's IT assets throughout their lifecycle. Many organizations continue to rely on spreadsheets or manual records to manage devices such as laptops, desktops, monitors, printers, networking equipment, and accessories, often resulting in duplicate records, data inconsistency, asset loss, and limited visibility into asset availability. AssetFlow addresses these challenges by providing a secure, AI-enabled platform for efficient IT asset lifecycle and inventory management.

The platform enables administrators and authorized users to register, categorize, assign, transfer, update, and monitor IT assets efficiently. Each asset is associated with essential information including asset ID, category, purchase details, assigned employee, department, location, condition, warranty information, and current status. To improve asset identification and tracking, every asset is assigned a unique **QR code** that provides quick access to asset information, assignment history, and lifecycle details. The system also maintains allocation, return, and maintenance records, while a centralized **analytics dashboard** provides real-time insights into asset distribution, inventory availability, and utilization. AI-powered analytics further assist in identifying maintenance requirements, warranty expiry trends, and asset replacement recommendations to support informed decision-making.

The frontend of AssetFlow is developed using **React.js**, providing a responsive and user-friendly interface. **Spring Boot 3** is used to develop the backend and RESTful APIs, while **Spring Security** and **JSON Web Tokens (JWT)** provide secure authentication and role-based access control. **PostgreSQL** serves as the relational database for securely storing asset, user, assignment, and inventory information.

To ensure consistent deployment across different environments, the application is containerized using **Docker**. **Git** and **GitHub** are used for version control and collaborative development, while **Jira** supports Agile software development through sprint planning, task management, issue tracking, and progress monitoring throughout the project lifecycle.

Overall, AssetFlow provides organizations with a secure, scalable, and intelligent solution for managing IT assets by improving asset visibility, simplifying lifecycle management, enhancing inventory accuracy, and enabling data-driven decision-making through modern software engineering practices.

Signature of the Guide HOD